BCA 303-4 Internet of Things Course Plan

Even Semester 2025

SECTION I

Semester	**IV**	Class	**4BCA A & B**
Course Code	**BCA303-4**	Course title	**Internet of Things**
Hours	**75**	Hours per week	**2+3**
Faculty name	**Dr Vandana Sharma**	Contact details	**Mobile: 9911143737** **Mail:** vandana.sharma@christuniversity.in
Class policies and guidelines	1. All communication will be done through CHRIST University Official Mail/official class WhatsApp group. 2. Read the course content / resources shared in Google Drive before coming to the class. 3. Usage of the Laptop is only with the prior permission (as per the requirements of the class/Lab/demo) 4. Help / Discussion / Mentoring will happen from 4:00 PM to 4:30 PM on weekdays, 12.30 PM to 1:00 PM on Saturday		
Course Description	This course covers the Introductory concepts associated with Internet of Things. This course aims to provide comprehensive knowledge of Internet of Things paradigms.		
Course Objectives	This course is an introduction into the basic concepts and principles of the Internet of Things. It outlines the different protocols for Internet of Things. The challenges and applications of IoT implementations are also discussed in detail in this course.		
Course Outcomes	Upon successful completion of this course, the students would be able to CO1: Understand the definition and significance of the Internet of Things. CO2: Demonstrate the ability to understand different protocols and architecture of the Internet of Things. CO3: Apply the knowledge to interface various sensors with IoT development board. CO4: Design and implement IoT system for real time applications.		

SECTION II					
Unit number and title	**Unit details**	**Week (starting and end dates)**	**Hours per Week**	**Learner Centered Approach**	**Resource/ Reference details**
Unit-1 Introduction to The Internet of Things	Evolution of IoT —IoT Definition – Elements of an IoT ecosystem Roles of Sensors & Actuators – Types of sensors – Working of Sensors	Week 1	5	Diagnostic Activity/ Activity: From Reels to Real Skills: IoT Learning Day	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	Development boards – Arduino – Raspberry pi – Beagle bone – ESP8266 – Interfacing of sensors with development boards	Week 2	5	Experiential Learning (Tinkercad)	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	IoT Networking Components – Addressing Strategies in IoT – Trends and implications – Operating System for IoT – Industrial IoT – IoT applications	Week 3	5	Problem-solving Sessions/ Activity: From Reels to Real Skills: IoT Learning Day **Assessment-1 (20 Marks**	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
Unit 1	Revision	Week 4	5		Text Books Ref. Books Online MOOC Content

					1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
Unit-2 **Sensors and Actuators**	Introduction – Sensors – Sensor Characteristics – Sensorial Deviations – Sensing Types – Scalar sensing – Multimedia sensing – Hybrid sensing – Virtual sensing	Week 5	5	Activity: Collaborative Learning	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	Sensing Considerations – Wearable sensors and their Applications Actuators	Week 6	5	Instructional activity: Brainstorming and problem solving	Text Books Ref. Books Online MOOC Content
	Actuator Types – Mechanical actuators – Hydraulic actuators – Thermal or magnetic – Electric actuators – Actuator Characteristics	Week 7	5	Formative Assessment: Quiz **Assessment-2 (20 Marks)**	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
Unit-3 Architecture and Communication Protocol	Layered Architecture for IoT – Protocol Architecture of IoT – IEEE 802.15.4 – Near Field Communication (NFC) – RFID	Week 8	5	Experiential Learning : Knowledge Review Session (Review by peer)	Text Books Ref. Books Online MOOC Content https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	ZigBee –Bluetooth Low Energy (BLE) – Long Term Evolution-	Week 9	5	Participatory Learning: Think-Pair-Share	Text Books Ref. Books

	Advanced			Activities	Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	Z-Wave – Components of ZWave Network – LoRa – NB-IoT – Wi-Fi	Week 10	5	Summative Assessment: **Concept Map (or) Mind map (Assessment-3: 10 Marks)**	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	Revision	Week 11	5		Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
Unit-4 IoT Communication Technologies	Introduction – Constrained nodes – Constrained networks – Types of constrained devices – Low power and lossy networks – Infrastructure Protocols – Internet protocol version 6 (IPv6) – RPL	Week 12	5	Diagnostic Assessment: Debugging Challenges **(Assessment-4: 30 Marks)**	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	MID SEMESTER EXAMINATION				

	IPv6 over Low Power Wireless Personal Area Networks (6LoWPAN) – Micro internet protocol (uIP) – Nano internet protocol (nanoIP) – Discovery Protocols	Week 13	5	Instructional Activity: Code Enrichment	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	Data Protocols – MQTT – CoAP – REST – Identification Protocols – Device Management – Semantic Protocols	Week 14	5	Participatory Learning: Role Play, Case Study	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
Unit-5 IoT Challenges	IoT Security Threats – IoT Security Requirements –Privacy in IoT Networks – Reliability Challenges – Interoperability of Devices – Reliability Issues Due to Energy Constraint	Week 15	5	Experiential Learning: Workshops and MooC courses	Text Books Ref. Books Online MOOC Content
	Cloud Platforms for IoT – Virtualization concepts and Cloud Architecture –	Week 16	5	Project Based learning /Lab Test	Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
	Cloud computing – Study of IoT Cloud platforms – ThingSpeak API and MQTT –	Week 17	5	Research Based Learning and Case Study **(Assessment-5: 20**	Text Books Ref. Books

	Interfacing ESP8266 with Web services			**Marks) Lab Test - 30 marks**	Online MOOC Content
All Units	Revision	Week 18	5		Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview
All Units	Revision	Week 19	5		Text Books Ref. Books Online MOOC Content 1. https://www.tinkercad.com/ 2. https://onlinecourses.nptel.ac.in/noc22_cs53/preview

LAB EXERCISES

UNIT – I	1. Study of Arduino UNO/ /Raspberry pi Board and Components using Tinkercad. 2. Blinking LED sketch with Arduino UNO/Raspberry pi using Tinkercad.	Week 2 Week 3
UNIT – II	3. Interfacing of Temperature sensor with Arduino/Raspberry Pi. Write a program to read the specific temperature of a room and display on the serial monitor. 4. Simulation of 4-Way Traffic Light with Arduino UNO/Raspberry pi	Week 4 Week 5 Week 6
UNIT – III	5. LED Fade Sketch and Button Sketch 6. Analog Input Sketch (Bar Graph with LEDs and Potentiometer)	Week 7 Week 8 Week 9
UNIT – IV	7. Digital Read Serial Sketch (Working with Motion/IR/Gas or Any other Sensor) 8. Spinning a DC Motor and Motor Speed Control Sketch	Week 10 Week 12 Week 13

UNIT -V	9. Interfacing Arduino UNO/Raspberry pi with Cloud (Thingspeak API) 10. Write a program to visualize data of temperature sensor on Arduino IoT cloud/Thingspeak	Week 14 Week 15 Week 16

Activity Description:

Unit	Activity Type	Description
Unit-1	Activity: Diagnostic Assessment	**Objective :** To understand the basics of IoT and analyze its features Activity : **Diagnostic assessment on "How Internet of Things are different from traditional networks?"** Students will form teams of five to discuss the features and advantages of IoT to other traditional networking methods. Creating an infographics and presenting it to other teams. Posting and tagging it on social media.
	Experiential Learning	**Objective:** To reinforce basic of IoT and Tinkercad. **Activity:** Use Tinkercad to understand the basics of IoT. Students will form teams and each team will discuss about IDE with basic setup and commands.
	Problem-solving Sessions.	The problem solving sessions and questions are designed to assess comprehension, application, and problem-solving skills, ensuring a comprehensive review of the material. Through this formative assessment, students receive immediate feedback on their performance, allowing them to identify areas of strength and areas needing improvement.
	Activity: From Reels to Real Skills: IoT Learning Day	In the first step, students are encouraged to form groups on their own. As a fun penalty, each group must create an Instagram reel before starting any group work. In Session-1, students will attend a tutorial session focused on IoT basics and take detailed notes. During Session-2, they will listen to the explanation of Exercise-1 and execute the exercise individually to practice the concepts learned. In Session-3, each group will present their Instagram reel, followed by a group presentation on their case study concerning the login strategy of their project. Session-4 involves attending a lecture on Tinkercad, with an emphasis on understanding these essential components of Arduino Uno. Finally, in Session-5, students will work on problems related to sensors and actuators, seeking guidance if needed to ensure thorough comprehension and skill application.
Unit 2	Participatory Learning: Poster Presentation	Divide the students into small groups of 3-4. Assign each group a specific sensor and actuators to focus on. Each group will create a poster with Abstract, Introduction, Procedure, Practical Uses Cases etc. Each group will present their poster to the class, explaining the key points and demonstrating examples.
	Instructional activity: Brainstorming	Divide students into small groups of 3-4. Provide each group with sticky notes or plain sheets. Ask each group to brainstorm and write down ideas for mini project. Each group can share their ideas with the class by explaining their thoughts. Based on the brainstormed ideas, collaboratively outline the design of the system.

	Formative Assessment: Quiz using Kahoot	The formative assessment using NearPod provides an engaging and interactive way to evaluate students' understanding of IoT concepts covered in the course. Students participate in a Game based quiz session where they answer multiple-choice questions related to IoT fundamentals, functions, and other key topics. Nearpod's platform allows for real-time feedback and instant scoring, fostering a competitive and motivating environment.
Unit 3	Experiential Learning : Code Review Session (Review by peer)	**Code Review Session**: Imagine you are working in a facility that tracks the data of the stored products. Each product will have multiple sensors. You need to write a function that tracks the data generated from each sensor and derive useful insights. **Question:** How does the sensors handle data? What is the threshold value for each sensor?
	Participatory Learning: Think-Pair-Share Activities	The Think-Pair-Share activity on sensors and actuators aimed to deepen students' understanding through active participation and peer collaboration. Initially, students individually reviewed the concept of IoT, including their definition, built-in methods, basic operations, and varieties of sensors and actuators. They then paired up to discuss their insights, share examples, and clarify misconceptions about IoT Applications. Through group discussion, students consolidated their learning by presenting key findings and exploring practical applications of Internet of Things. **Scenario:** Consider a scenario where you simulate an application of IoT using Arduino UNO. . Write a code to simulate IoT application and display data on dashboard in real time.
	Summative Assessment: Concept Map (or) Mind map	The summative assessment activity involved students creating a comprehensive concept map or mind map on IoT, with a particular emphasis on sensors. This visual representation required students to organize and interconnect various methods and concepts of IoT. The focus on IoT protocols encouraged students to delve deeper into their functionality, exploring common operations, and practical applications. This assessment not only reinforced students' understanding of IoT but also enhanced their ability to visually synthesize and communicate complex information, fostering a deeper grasp of the subject matter.
	Diagnostic Assessment: Debugging Challenges	The diagnostic assessment activity involved a debugging challenge focused on revising IoT concepts and comprehensions, including Arduino Uno and popular IoT Protocols. Students were tasked with identifying and fixing errors in provided code snippets, each designed to test their understanding of these concepts. This hands-on activity not only reinforced their knowledge of IoT but also honed their debugging skills, critical thinking, and problem-solving abilities. By engaging in this challenge, students gained practical experience in correcting code and

		deepened their comprehension of real time data management techniques using Tinker cad.
Unit 4	**Instructional activity - Code Enrichment**	The instructional activity on Code Enrichment centered around the topic of IoT (Unit I, II and III). Students were guided through the process of writing and enhancing IoT functions. This activity provided hands-on experience in creating real time IoT Applications. By engaging in this activity, students enhanced their innovative skills, gained a deeper understanding of IoT, and learned best practices for IoT applications.
	Participatory Learning - Role play	In the participatory learning activity, students engaged in a role-play exercise focusing on the interfacing of Arduino Uno. Additionally, the role-play included the implementation and execution of functions, where students acted out the step-by-step simulation. This interactive approach fostered a collaborative learning environment, allowing students to grasp complex concepts through practical application and peer interaction.
Unit 5	**Experiential Learning: Workshops**	In the experiential learning activity, students participated in a workshop led by a top industry professional, providing them with invaluable insights into the practical applications of IoT. The expert shared real-world experiences and demonstrated advanced techniques, bridging the gap between academic concepts and industry practices. This hands-on workshop allowed students to engage directly with cutting-edge tools and methodologies, enhancing their understanding of how IoT is used in various Industry. The interaction with a seasoned professional also offered networking opportunities and inspired students to envision potential career paths in the tech industry.
	Experiential Learning: PBL (Project Deployment)	In this experiential learning activity, students engaged in project-based learning with a focus on project deployment, culminating in an industry-focused presentation. Throughout the project, students applied their IoT skills to develop and deploy real-world applications, gaining hands-on experience with tools and methodologies used in the industry. The final presentation required students to showcase their projects to industry professionals, receiving feedback and insights on their work. This experience not only enhanced their technical skills but also improved their ability to communicate complex concepts effectively, bridging the gap between academic learning and industry expectations.
	Research Based Learning and Case Study	In this research-based learning activity, students conducted in-depth investigations on specific IoT topics, culminating in a detailed case study. They explored various aspects of IoT application development, through rigorous research and practical experimentation. The case study required them to analyze real-world scenarios, apply their findings, and document their methodologies and results. This approach not only deepened their understanding of IoT but also honed their analytical and problem-solving skills, preparing them for complex challenges in both academic and professional settings.

SECTION III

Assessment outline:

This course will follow the assessment outline given below:

CIA - I	CIA – II	Attendance	ESE I	ESE II	ESE III
40	50	10	20	30	50
CIA 100 marks will be converted to 50 marks CIA-I will be considered as **Assessment-1:** Problem-solving Sessions **Assessment-2:** Formative Assessment (Certifications and Case Study) CIA-II **Regular Lab Evaluation (50 Marks)** **Attendance marks: 10 Marks**			ESE 100 marks will be converted to 50 marks ESE-I: **Assessment-3 - Summative Assessment: Concept Map (or) Mind map (20 Marks)** ESE-II: **Assessment-4:** Presentation **(30 Marks)** ESE-III: **Group Project: 50 Marks**		

Mapping: A template to map the Learning Outcomes of the course against the components of assessment is given below:

Course Outcomes	Components of assessment			
	LRNG	**Cross-cutting issues relating to Professional Ethics, Gender, Human Values, Environment and Sustainability**	**Employability, Entrepreneurship/ Skill Development**	**Sustainable Development Goals**
CO 1	Global	Environment and Sustainability	Employability and Skill Development	SDG 4, 17
CO 2	Global	Environment and Sustainability	Employability and Skill Development	SDG 4, 17
CO 3	Global	Environment and Sustainability	Employability and Skill Development	SDG 4, 17

SECTION IV

Assessment Description: CIA - I

Assessment Name	CIA I (A and B)
Schedule	18th January
Weightage	40

Individual Assignment Details	Individual
Assessment description:	**Assessment-1** Certificate Corse **from NetAcad.com (20 Marks) + Assessment-2** Case Study **(20 Marks)**
Mode of the Test	Assessment 1- Online, Assessment 2- Classroom
Marks	20 + 20 Marks
Duration	1 Hour + 1 Hour
Topics Covered	Unit-1, Unit-2
Remarks	● No Repeat Test ● Timely Submission

Evaluation rubrics:

Quiz Score	Impression	Problem-solving Sessions:	Impression
Above 60%	Proficient	Above 60%	Proficient
40-59%	Satisfactory	40-59%	Satisfactory
0- 39%	Need to improve	0- 39%	Need to improve

Mapping the Learning Outcomes with components of the evaluation rubrics:

Learning Outcomes	Method of assessment	Component of the evaluation rubrics
IoT Fundamental concepts	Problem solving sessions/ Viva Voce	✓ Impression

Assessment Description: CIA – II

Assessment Name	CIA -II
Schedule	Week-12
Weightage	50%
Individual Assignment Details	Individual
Assessment description:	Diagnostic Assessment: Regular Lab Based Assessments **(50 Marks)**
Mode of the Test	Practical Submission
Marks	50 Marks
Duration	2 Hours
Topics Covered	Unit-1,2,3,4,5

Remarks	

Evaluation Rubrics:

S. No.	Evaluation Rubrics (Debugging Challenge)	Criteria	Marks
1	Identification of Bugs	Excellent	90-100%
2	Correction of Bugs	Good	70-89%
3	Use of Debugging Tools and Techniques	Satisfactory	50-69%
4	Documentation and Explanation	Needs Improvement	30-49%
5	Problem-Solving Skills	Unsatisfactory	0-29%

S. No.	Evaluation Rubrics (Case study)	Marks	Remarks
1	Understanding and Analysis	10%	Excellent 90-100% Good 70-89% Satisfactory 50-69% Needs Improvement 30-49% Unsatisfactory
2	Research and Use of Evidence	10%	
3	Organization and Structure	10%	
4	Clarity and Precision	10%	
5	Creativity and Originality	10%	

Assessment Description: **ESE – I**

Assessment Name	ESE -I
Schedule	Week-9
Weightage	20%
Individual Assignment Details	Individual
Assessment description:	**Assessment-3 - Summative Assessment:** **Concept Map (or) Mind map (20 Marks)**
Mode of the Test	Activity Task
Marks	20 Marks
Duration	1 Hour

Topics Covered	Unit-3
Remarks	

Evaluation Rubrics:

S. No.	Evaluation Rubrics (Concept map/ Mind map)	Criteria	Marks
1	Content Coverage (6 Marks)	Excellent	90-100%
2	Organization and Structure (6 Marks)	Good	70-89%
3	Visual Design and Creativity (4 Marks)	Satisfactory	50-69%
4	Clarity and Readability (2 Marks)	Needs Improvement	30-49%
5	Depth of Analysis (2 Marks)	Unsatisfactory	0-29%

Assessment Description: ESE – II

Assessment Name	ESE II
Schedule	Week 12
Weightage	50%
Individual Assignment Details	Individual
Assessment description:	Presentation
Mode of the Test	Online Submission and Viva Voce
Marks	30 Marks
Duration	Class Hours
Topics Covered	Unit I to V
Remarks	● No Late Submission ● GCR submissions are mandatory.

Evaluation Rubrics:

S. No.	Evaluation Rubrics	Marks
1	On-time completion	1
2	Formatting & Presentation	1
3	Clarity of the topic	1

4	Domain based knowledge	1
5	Viva	1

Assessment Description: ESE – III

Assessment Name	ESE –III
Schedule	Week-14
Weightage	30%
Individual Assignment Details	Group Project
Assessment description:	Group Project Demonstration and Presentation / Lab Test
Mode of the Test	Project Presentation / Lab Test
Marks	50
Duration	3 Hours
Topics Covered	Unit 1-5
Remarks	• Project will be starting with Project domain selection, Synopsis submission, developed and deployed according to one of the software process models.

Evaluation Rubrics for project:

Evaluation rubrics

Component	Max Marks	Need to improve	Satisfactory	Proficient
R1: Conceptual Clarity	10	Vague understanding or major conceptual gaps (0–4)	Adequate understanding with minor gaps (5-7)	Clear, comprehensive understanding of data science concepts and tools (8-10)
R2: Application & Analysis	10	Unable to apply concepts or shows incorrect analysis (0–4)	Partial application with some analytical errors (5–7)	Accurate application of methods with proper data analysis and interpretation (8-10)
R3: Problem Solving	10	Incorrect approach or incomplete solution (0–4)	Partial or partially correct solution with minor logical gaps (5–7)	Complete, well-reasoned solution demonstrating critical thinking (8–10)
R4: Presentation	10	Poor organization, unclear visuals or documentation (0–4)	Moderately structured presentation; some clarity issues (5–7)	Well-structured, clear, and professional presentation of

				project outcomes (8–10)
R5: Mapping to SDG	10	Vague understanding or major conceptual gaps (0–4)	Adequate understanding with minor gaps (5-7)	Clear, comprehensive understanding of data science concepts and tools (8-10)

Evaluation Rubrics for Lab Test:

Score	**Impression**
>90%	Proficient
80-90%	Very Good
60-80%	Good
40-60%	Average
< 40%	Need to improve

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